

Manufacturing Quality Comparison Report

F-Test Statistical Analysis

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Executive Summary

Conclusion: FAIL TO REJECT NULL HYPOTHESIS
The two plants have SIMILAR variance levels (quality consistency) at 5.0% significance level.
Plant with Better Consistency: Plant 1 (32.43% lower variance)
P-Value: 0.568527

Statistical Results

Metric	Plant 1	Plant 2
Sample Size (n)	10	10
Mean	20.000000	20.040000
Variance (s²)	0.033333	0.049333
Std. Deviation (s)	0.182574	0.222111

F-Test Results

Parameter	Value
F-Statistic	1.480000
Degrees of Freedom (numerator)	9
Degrees of Freedom (denominator)	9
P-Value (two-tailed)	0.568527
Significance Level (α)	0.05
Test Conclusion	FAIL TO REJECT NULL HYPOTHESIS

Interpretation

Null Hypothesis (H₀): The variances of Plant 1 and Plant 2 are equal ($\sigma_1^2 = \sigma_2^2$)
Alternative Hypothesis (H₁): The variances are not equal ($\sigma_1^2 \neq \sigma_2^2$)
Test Result: The two plants have SIMILAR variance levels (quality consistency) at $\alpha = 0.05$ significance level.

Plant with Better Consistency: Plant 1 has lower variance, indicating higher consistency in quality measurements (32.43% improvement).